

Compute-Selected Generated-Video Content Pockets Predict Delayed Human Recognition

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Abstract

Brain-predictive and self-supervised video models can rank generated videos, but proxy scores do not by themselves establish that people remember the selected clips. We test a narrow prediction from a Stable Video Diffusion replay regime: two content pockets, orange flowers and hanging clothes, were first stabilized by TRIBE/BMD replay residuals and exact V-JEPA verifier geometry against hard negative controls. We then froze a two-session old-vs-lure recognition task and ran a delayed Prolific study. In the complete Wave 2 sample, 62 participants submitted 25 forced-choice recognition trials each. After excluding media-error-flagged trials, primary positive pockets were recognized on 114/123 trials (92.7%, Wilson 95% CI [86.7%, 96.1%]), compared with 150/186 hard negative control trials (80.6%, [74.4%, 85.7%]). The participant-level primary-positive minus hard-control contrast was +11.7 percentage points (bootstrap 95% CI [+4.4, +19.4], sign-flip $p = 0.00425$). Hanging clothes was individually robust; orange flowers was high in absolute recognition but weaker as a standalone contrast. These results support a narrow claim: compute-selected generated-video content pockets can predict delayed human old-vs-lure recognition advantage. They do not establish broad human memorability, measured-fMRI/BMD grounding, prompt-conditioned generation control, or a general black-box optimization result.

1 Introduction

Video generators make it increasingly cheap to sample many plausible clips for a single creative intent. The bottleneck then shifts from generation to selection: which generated clips are likely to matter to a viewer? Standard video generation metrics emphasize visual quality, temporal consistency, text-video alignment, or broad preference [1, 2]. Those metrics are necessary, but they do not directly ask whether viewers will remember the clip.

Human memorability is partly stable across observers and can be predicted from visual and semantic representations [3–5]. Brain-aligned video representations and neural-response prediction models provide another candidate frame: generated clips can be scored by predicted cortical response, self-supervised video embeddings, and related descriptor geometry [6–8]. However, a score in a proxy model is not a human memory result. A generated-video candidate selected by TRIBE, V-JEPA, CLIP, or any other model must still clear a behavioral gate before the word “memorability” is used strongly.

This preprint isolates one such gate. Earlier compute-side audits in our image-conditioned Stable Video Diffusion (SVD) replay regime [9] found that replay residuals were dominated by seed-image/content identity rather than by alpha/guidance recipe choice. Orange flowers and hanging clothes emerged as the strongest non-jellyfish positive content pockets. Aerial beach, city street,

and storm beach served as hard negative controls. Descriptor-conditioned replication showed that the primary orange/hanging pockets transported under fresh stochastic seeds with positive TRIBE scores and exact V-JEPA support, while the hard controls stayed negative. Generated-video CLIP did not pass prospectively, so the validation packet was explicitly V-JEPA-caveated.

The present contribution is intentionally modest:

1. We translate a compute-proxy content-pocket finding into a frozen two-session delayed recognition-memory task.
2. We use same-category lures and hard negative controls so the task is not merely category familiarity.
3. We show that the pooled primary content-pocket packet is recognized more accurately than hard controls.
4. We keep the claim narrow: delayed old-vs-lure recognition for this generated-video packet, not broad memorability or generator steering.

2 Related Work

Visual and video memorability. Image memorability work showed that some images are consistently remembered better than others across observers [3]. Video memorability extends this to dynamic clips, semantic events, and temporal decay [4, 5]. The MediaEval video memorability tasks further separate short-term memorability, long-term memorability, dataset generalization, and EEG-adjacent prediction subtasks [10]. These results motivate behavioral validation, but they do not remove the need to test generated-video stimuli directly.

Brain-aligned and self-supervised video representations. BOLD Moments links short naturalistic videos, fMRI responses, and memorability metadata [6]. TRIBE v2 is a multimodal brain-encoding model for predicting cortical responses from vision, audition, and language [7]. V-JEPA-style self-supervised video models provide a strong non-brain representation frame for video understanding [8]. In the current study, TRIBE and V-JEPA are used to select and verify candidate pockets, not as substitutes for human behavior.

Brain-to-media and response-guided generation. Recent fMRI-to-image and fMRI-to-video systems use diffusion models, CLIP-style conditions, semantic guidance, and temporal reconstruction modules to reconstruct perceived content from brain activity [11–15]. Conversely, response-guided and saliency-guided generation methods use neural or attention predictors to steer image synthesis [16–18]. Our setting is different: we do not decode media from measured brain activity, and we do not claim a differentiable neural reward for video generation. We test whether a compute-selected generated-video content pocket survives delayed human recognition.

3 Compute-Side Candidate Regime

3.1 Artifact regime

The candidate pool consists of seed images, image-conditioned SVD generations, fixed recipe neighborhoods, visual-gate statuses, TRIBE/BMD replay scores, exact V-JEPA video features, CLIP

Table 1: Analysis arms in the delayed old-vs-lure recognition task.

Arm	Group	Role
Orange flowers	Primary positive	TRIBE/V-JEPA-verified candidate pocket
Hanging clothes	Primary positive	TRIBE/V-JEPA-verified candidate pocket
Aerial beach	Hard negative control	Matched negative content pocket
City street	Hard negative control	Matched negative content pocket
Storm beach	Hard negative control	Matched negative content pocket

diagnostics, and pocket labels. Prompt text is metadata-only in the current SVD replay runner, so prompt rewriting is not an active image-generation intervention in this regime.

3.2 Primary and control pockets

The primary positive pockets are orange flowers and hanging clothes. The hard negative controls are aerial beach, city street, and storm beach. Blue jellyfish and old car were retained as secondary boundary pockets in the broader research ledger, but they are not part of the primary human-recognition claim here.

3.3 Prospective verifier boundary

Fresh descriptor-conditioned replication generated and scored 90/90 clips with zero visual failures. Orange flowers and hanging clothes stayed positive under TRIBE; hard controls stayed negative. Exact V-JEPA transported prospectively, but generated-video CLIP did not. The human validation packet was therefore allowed to use orange/hanging as TRIBE/V-JEPA-verified primary pockets, while not claiming a two-descriptor V-JEPA+CLIP prospective pass.

4 Human Recognition-Memory Study

4.1 Stimuli

Each participant saw one old target from each analysis arm in Session 1. Session 2 presented each old target against a newly generated same-category lure. Sparse forms prevented any participant from seeing multiple old targets from the same analysis arm. Fillers were included to stabilize the cover task and reduce the salience of the five analysis arms.

4.2 Procedure

Session 1 was a viewing and cover-task session. Session 2 was a delayed two-alternative forced-choice recognition task: participants selected which clip they had seen in Session 1. Participant identifiers were used only for study matching/payment and were reduced to aggregate counts in committed analysis artifacts. The endpoint is exact old-vs-lure recognition accuracy, which is stronger than preference but narrower than free recall, long-term memory, or measured neural validation.

4.3 Data integrity

The Wave 2 Prolific export contained 64 rows: 60 approved, 2 awaiting review, and 2 timed out. The webhook export contained complete real Session 2 payloads for all 62 completed submissions

and none for the two timed-out submissions. There were no duplicate real Prolific IDs in complete Session 2 payloads.

The Wave 1 webhook inbox was capped before a paid upgrade, so the full Session 1 JSON payload set is not available. The retained Wave 1 subset contains 24 complete payloads; 21 overlap with complete Wave 2 participants, and all 21 have matching deterministic form assignments. We therefore report the result as Prolific-confirmed Wave 1 exposure plus complete Session 2 recognition, with partial Wave 1 JSON retention as a provenance caveat.

4.4 Analysis

Primary analyses exclude trials flagged with media errors. We report Wilson confidence intervals for binomial recognition accuracy, exact binomial tests against chance, and participant-level paired contrasts between the primary positive packet and the hard-control pool. Paired uncertainty is estimated with bootstrap resampling over participants. A sign-flip permutation test evaluates whether the paired primary-minus-hard contrast differs from zero.

5 Results

5.1 Recognition accuracy

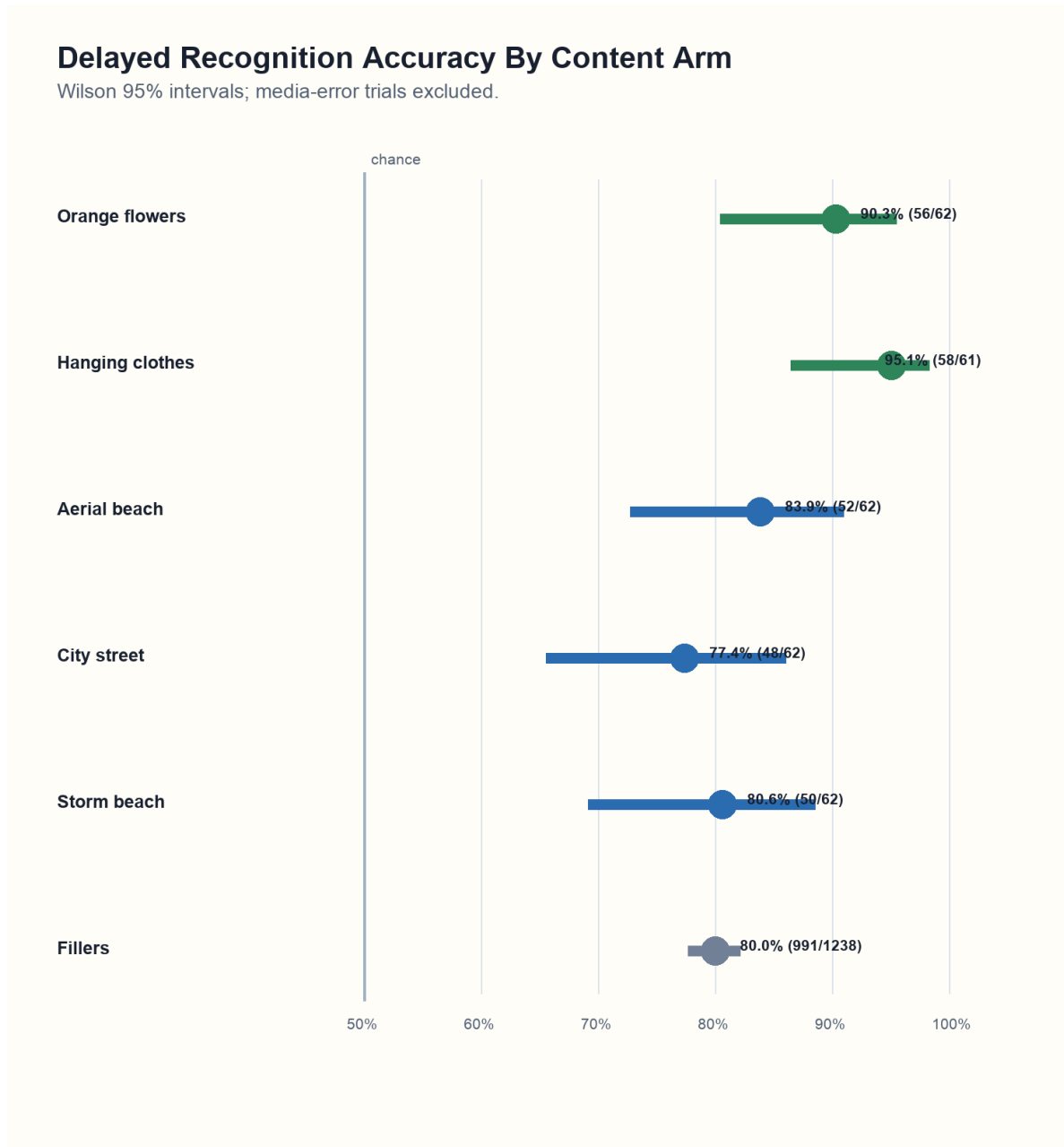


Figure 1: Delayed old-vs-lure recognition accuracy by content arm after excluding media-error-flagged trials. Points show accuracy and horizontal bars show Wilson 95% confidence intervals.

The hard negative controls were also recognized above chance. The result is therefore not that only the positive pockets are memorable. The supported claim is relative: the primary positive packet was recognized more accurately than the hard-control pool under the same sparse old-vs-lure task.

Table 2: No-media-error recognition results.

Group	Correct / n	Accuracy	Wilson 95% CI	Exact p vs 0.5
Primary positives	114 / 123	0.927	[0.867, 0.961]	2.68×10^{-24}
Hard negative controls	150 / 186	0.806	[0.744, 0.857]	9.63×10^{-18}
Unrelated fillers	991 / 1238	0.800	[0.777, 0.822]	7.69×10^{-106}
Orange flowers	56 / 62	0.903	[0.805, 0.955]	2.97×10^{-11}
Hanging clothes	58 / 61	0.951	[0.865, 0.983]	3.29×10^{-14}
Aerial beach	52 / 62	0.839	[0.728, 0.910]	5.71×10^{-8}
City street	48 / 62	0.774	[0.656, 0.860]	1.74×10^{-5}
Storm beach	50 / 62	0.806	[0.691, 0.886]	1.21×10^{-6}

5.2 Primary packet contrast

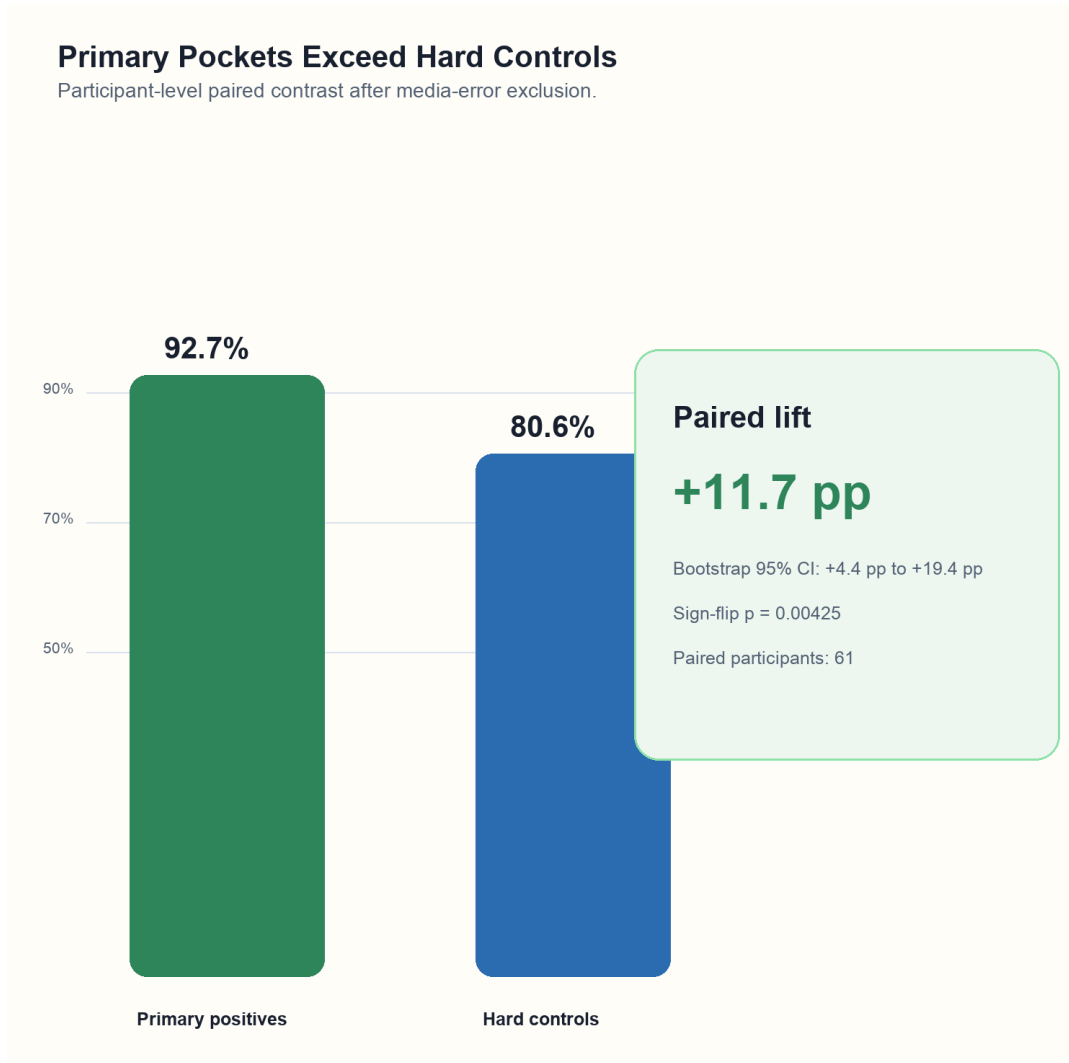


Figure 2: Pooled primary-positive packet versus hard controls. The paired participant-level lift was +11.7 percentage points after excluding media-error-flagged trials.

The primary-positive minus hard-control contrast was +11.7 percentage points (bootstrap 95% CI [+4.4, +19.4], sign-flip $p = 0.00425$) across 61 complete paired participants after media-error exclusion. The all-trial sensitivity analysis was qualitatively similar: +12.1 percentage points, bootstrap 95% CI [+4.6, +19.9], sign-flip $p = 0.00287$, across 62 complete paired participants.

5.3 Individual pocket contrasts

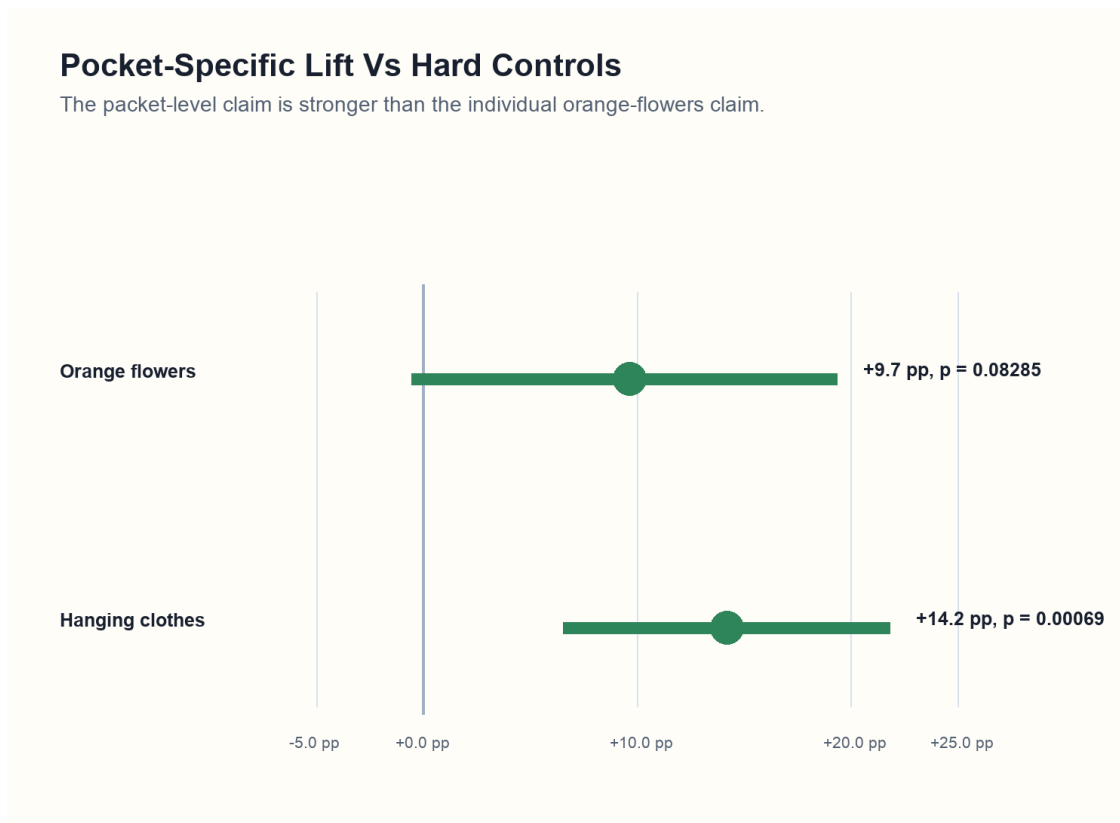


Figure 3: Pocket-specific paired lift versus the hard-control pool. Hanging clothes clears the individual contrast; orange flowers is high in absolute recognition but weaker as a standalone hard-control contrast in this wave.

Hanging clothes exceeded the hard-control pool by +14.2 percentage points (bootstrap 95% CI [+6.6, +21.9], sign-flip $p = 0.00069$). Orange flowers was high in absolute recognition but weaker as a standalone contrast: +9.7 percentage points, bootstrap 95% CI [-0.5, +19.4], sign-flip $p = 0.08285$. The honest claim is therefore the pooled primary packet, with pocket-specific nuance preserved.

6 Discussion

This study upgrades one content-pocket line from compute-proxy evidence to a narrow human behavioral result. The selection path was prospective at the claim-relevant level:

TRIBE/V-JEPA content-pocket candidates
→ frozen old-vs-lure stimuli
→ delayed human recognition endpoint.

The result does not imply that any arbitrary TRIBE-high generated video is more memorable to humans. It says that, within this SVD content-pocket packet, the primary positive pockets selected by the compute regime were later recognized more accurately than hard controls. Content identity can therefore be a real memorability-bearing axis in generated-video candidate selection, and future selector studies should preserve content-pocket controls, same-category lures, and independent human endpoints.

7 Limitations

First, the original full response-analysis plan named a larger minimum usable sample. This Wave 2 result is strong enough to draft around, but it should not be presented as a large-sample final confirmation.

Second, the full Wave 1 webhook payload set was not retained because the inbox was capped before upgrade. Prolific completion and deterministic form assignment support the exposure record, and the retained Wave 1 subset shows zero form mismatches, but complete Session 1 JSON provenance is absent.

Third, the task measures exact two-alternative recognition against same-category lures. It does not measure free recall, long-term memory, attention, engagement, preference, quality, emotion, virality, or commercial outcomes.

Fourth, the positive pockets are content categories. This result does not prove that alpha/guidance recipes, black-box optimization, or prompt text caused memorability improvements. In the current SVD runner, prompt text is metadata for generation.

Fifth, BMD and measured-fMRI grounding remain separate gates. TRIBE/V-JEPA helped choose the packet, but this human study is behavioral recognition evidence, not measured neural validation.

8 Ethics and Data Availability

Participants were recruited through Prolific for a minimal-risk online recognition task involving short generated videos. The committed analysis artifacts contain aggregate counts and summary statistics only. Raw Prolific/webhook exports are excluded from the repository because they contain participant metadata. The reproducible aggregate analysis script and participant-safe result JSON are included with the source release.

9 Conclusion

Compute-selected generated-video content pockets can predict a delayed human old-vs-lure recognition advantage in this packet. The result is narrow but important: it gives the content-pocket regime a behavioral foothold while preserving the distinction between proxy scoring, recognition memory, measured brain validation, and broad memorability claims.

A Claim Contract

Allowed now:

- The primary orange/hanging content-pocket packet has delayed human old-vs-lure recognition evidence.

- Pooled primary positives exceed hard controls in participant-level recognition accuracy.
- Hanging clothes is individually robust in this wave.
- Orange flowers has high absolute recognition and supports the pooled primary result, but is weaker as a standalone hard-control contrast.

Not allowed:

- Human memorability is solved.
- BO-generated or TRIBE-selected videos are broadly more memorable to humans.
- TRIBE/V-JEPA scores are human memory.
- Prompt rewriting controls memorability in the current SVD runner.
- The result is measured-BMD or fMRI validation.

B Aggregate Provenance

The aggregate result is generated by `scripts/summarize_content_pocket_recognition_wave2.py`. Committed result artifacts are:

- `content_pocket_recognition_response_analysis_result_20260612.json`
- `content_pocket_recognition_response_analysis_result_20260612.md`
- `figures/content_pocket_recognition_accuracy_20260612.svg`
- `figures/content_pocket_recognition_contrast_20260612.svg`
- `figures/content_pocket_recognition_arm_contrasts_20260612.svg`

References

- [1] Ziqi Huang et al. Vbench: Comprehensive benchmark suite for video generative models. arXiv preprint arXiv:2311.17982, 2023.
- [2] Yaofang Liu et al. Fetv: A benchmark for fine-grained evaluation of open-domain text-to-video generation. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [3] Phillip Isola, Devi Parikh, Antonio Torralba, and Aude Oliva. Understanding the intrinsic memorability of images. In *Advances in Neural Information Processing Systems*, 2011.
- [4] Romain Cohendet, Claire-Helene Demarty, Ngoc Q. K. Duong, and Martin Engilberge. Videomem: Constructing, analyzing, predicting short-term and long-term video memorability. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.
- [5] Anelise Newman, Camilo Fosco, Vincent Casser, Allen Lee, Barry McNamara, and Aude Oliva. Multimodal memorability: Modeling effects of semantics and decay on video memorability. In *European Conference on Computer Vision*, 2020.

- [6] Benjamin Lahner et al. Modeling short visual events through the bold moments video fmri dataset and metadata. *Nature Communications*, 2024.
- [7] Stephane d’Ascoli, Jeremy Rapin, Yohann Benchetrit, Teon Brooks, Katelyn Begany, Josephine Raugel, Hubert Banville, and Jean-Remi King. A foundation model of vision, audition, and language for in-silico neuroscience. arXiv preprint arXiv:2605.04326, 2026.
- [8] Adrien Bardes et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. arXiv preprint, 2025.
- [9] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, Varun Jampani, and Robin Rombach. Stable video diffusion: Scaling latent video diffusion models to large datasets. arXiv preprint arXiv:2311.15127, 2023.
- [10] Lorin Sweeney, Mihai Gabriel Constantin, Claire-Helene Demarty, Camilo Fosco, Alba G. Seco de Herrera, Sebastian Halder, Graham Healy, Bogdan Ionescu, Ana Matran-Fernandez, Alan F. Smeaton, and Mushfika Sultana. Overview of the mediaeval 2022 predicting video memorability task. In *MediaEval Multimedia Benchmark Workshop Working Notes*, 2022.
- [11] Yizhuo Lu, Changde Du, Dianpeng Wang, and Huiguang He. Minddiffuser: Controlled image reconstruction from human brain activity with semantic and structural diffusion. arXiv preprint arXiv:2303.14139, 2023.
- [12] Zijiao Chen, Jiaxin Qing, and Juan Helen Zhou. Cinematic mindscapes: High-quality video reconstruction from brain activity. arXiv preprint arXiv:2305.11675, 2023.
- [13] Zixuan Gong, Guangyin Bao, Qi Zhang, Zhongwei Wan, Duoqian Miao, Shoujin Wang, Lei Zhu, Changwei Wang, Rongtao Xu, Liang Hu, Ke Liu, and Yu Zhang. Neuroclips: Towards high-fidelity and smooth fmri-to-video reconstruction. In *Advances in Neural Information Processing Systems*, 2024.
- [14] Jingyuan Sun, Mingxiao Li, Zijiao Chen, and Marie-Francine Moens. Neurocine: Decoding vivid video sequences from human brain activities. arXiv preprint arXiv:2402.01590, 2024.
- [15] Minghan Yang, Lan Yang, Ke Li, Honggang Zhang, Kaiyue Pang, and Yizhe Song. Semvideo: Reconstructs what you watch from brain activity via hierarchical semantic guidance. arXiv preprint arXiv:2602.21819, 2026.
- [16] Pawel Pierzchlewicz, Konstantin Willeke, Arne Nix, Pavithra Elumalai, Kelli Restivo, Cate Nealley, Gabrielle Rodriguez, Saumil Patel, Katrin Franke, Andreas Tolias, and Fabian H. Sinz. Energy guided diffusion for generating neurally exciting images. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [17] Yunxiang Zhang, Nan Wu, Connor Z. Lin, Gordon Wetzstein, and Qi Sun. Gazefusion: Saliency-guided image generation. *ACM Transactions on Applied Perception*, 2025. doi: 10.1145/3694969.
- [18] Xiwen Wang, Jizhe Zhou, Xuekang Zhu, Cheng Li, and Mao Li. Saliency guided optimization of diffusion latents. arXiv preprint arXiv:2410.10257, 2024.